

Crew Scheduling Optimization for Airline Operations

1. Model Formulation

Decision Variable

- $\text{Assign}(c, f)$: equals 1 if crew member c is assigned to flight f , otherwise 0.

Objective

Minimize the total cost of crew assignments based on flight hours and hourly rates.

Constraints

- **Coverage:** Each flight must have 1 pilot and 2 cabin crew
- **Qualification:** Crew must be certified for the aircraft type
- **Workload:**
 - Pilots ≤ 100 hours/month
 - Cabin crew ≤ 150 hours/month
- **Rest Time:** Minimum 24 hours between assigned flights

2. Task 1 Results

The model successfully assigned crew to all flights while satisfying all constraints and minimizing total cost. This serves as the baseline solution.

3. Task 2: New Pilot Regulations

New rules:

- Rest time increased to 48 hours
- Maximum flying hours reduced to 50 hours/month

Impact:

- Pilots can take fewer flights
- Scheduling becomes more restricted
- The total pilot cost increases due to reduced flexibility

4. Task 3: Revenue Maximization

To keep pilot cost the same as Task 1, not all flights can be operated.

New Variable

- Operate(f): equals 1 if a flight is operated, otherwise 0

Changes

- Crew can only be assigned if the flight is operated
- Pilot cost is limited to the original budget
- Objective becomes maximizing total revenue

Result:

- Flights with lower revenue or higher cost are cancelled
- The model keeps the most profitable flights
- Revenue loss is calculated by comparing achieved revenue to maximum possible revenue

5. Task 4: Crew Vacations

- A crew member cannot be assigned to flights that overlap with their vacation
- We define availability for each crew member and flight
- If a flight overlaps with a vacation, assignment is not allowed

Impact:

- Reduces available crew options
- May increase cost or limit scheduling
- Easy to integrate into the model